

**PUBPOL 4390/5390
PROBLEM SET 1****Due 11:55pm, Monday September 29, via Canvas**

- Work in teams of 1-3. You can choose your own teams. Clearly indicate all the members of the team at the beginning of the problem set.
- Only one submission to Canvas per team, please.
- Turn in the main problem set as one file only, not several documents. PDF preferred. Clearly label the problem set file (example: “PS 1 – PADM 5390 – Olivero Miller.pdf”).
- Please also upload, as auxiliary files, the programs used in your computations (eg Stata’s .do files).

1. Predicting Birthweight

Use dataset E to train OLS models predicting birthweight, and when you are satisfied, use dataset F as an out-of-sample dataset to assess your models’ performance. Important – don’t use the results in dataset F to choose among candidate models; instead just use dataset F once at the end, to measure goodness of fit, after you have committed to your models.

I would like you to develop 3 different OLS models, based on the following principles:

Model #		# of covariates	
1	Small	$K \leq 10$	Keep it simple, but still try to get good predictions.
2	Moderate	$20 < K < 50$	
3	Large	$200 < K$	Include polynomial or spline transformations, rich interactions between variables, etc. Try to build a model rich enough to run the risk of over-fitting, even with this larger dataset!

Note: the data (both “training” and “testing”) have not been pre-cleaned – this is different than the data we used in our in-class exercises.

- In your write-up for this problem, summarize (e.g. in a bullet point list) the data cleaning and checking steps that you undertook.
- Briefly describe each of your three models.
- Describe the in-sample (Dataset E) and out-of-sample (Dataset F) goodness of fit
 - Use both R^2 and RMSE. For RMSE, use the “ $1/N$ ” formula. Make sure to also show each sample’s raw standard deviation of birthweight, as a baseline to compare the RMSE against.
- For each model: how good a job does the “ $1/(N-K)$ ” RMSE formula do, when used on the training sample, at predicting the testing sample RMSE?
- If you were to use this exercise as a teaching moment for a friend or colleague who is not taking our class, what are the key lessons from this exercise that you would want to pass along to them?

2. Nonparametric Regression

For this problem, download the stata dataset “kenya.dta”. The data are from a project that Michael Kremer, Ted Miguel, and Rebecca Thornton did in rural Kenya. The final version of the paper is at:

<http://www.mitpressjournals.org/doi/abs/10.1162/rest.91.3.437>

This problem examines the relationship between merit awards and academic performance, as measured by school exams, among Kenyan schoolchildren. In early 2001, Grade 6 girls in a random subset of “treatment” schools (variable name “treat”) were offered a large cash award if they scored in the top 15% of all treatment school girls. These girls are cohort 1 (indicator variable “c1”). There is also a cohort 2, who were grade 6 girls who faced the incentive in 2002. (We will focus on “c1”.) The winners in each cohort are given by the binary variables (“winn01” “winn02”)

The dataset contains test score information from late 2000, the year before the program, late 2001, the first year of the program, and 2002, the second year of the program. The test score outcomes (“test00”, “test01” “test02”) were normalized such that the test distribution in the comparison schools is mean zero with a standard deviation of one (for all students in that grade, not just those in this sample – thus the mean need not equal zero in the sample).

The goal of this exercise is to estimate overall program treatment effects using non-parametric methods. An important question for the analysis is whether girls at the bottom of the baseline test score distribution were harmed by the program – perhaps due to demoralization or diversion of teacher attention to high-achieving classmates.

For this problem, use the stata dataset “kenya.dta”. You will only need the “cohort 1” data.

3a) Perform a locally-weighted linear non-parametric regression on the specifications listed below. (I think the Stata command `lpoly`, for “local polynomial”, is probably a good function to use here. If so, make sure to use the right option to set polynomial order to 1, for “linear”.) Use your own judgment (and eyeballs) to pick a good bandwidth and kernel.

Estimate the relationship on the non-extreme values of the 2000 distribution, only plotting the 2000 test score interval of $[-1.3, 1.6]$, for simplicity (you can use data outside of this range, as inputs into estimating the models).

- (i) Treatment girls, (nonparametrically) regress the 2001 test (dependent variable) on the 2000 test.
- (ii) Comparison girls, (nonparametrically) regress the 2001 test (dependent variable) on the 2000 test.
- (iii) Try to replicate the “treatment effect” line from Figure 3, Panel A of the published article. (for now, don’t worry about the confidence interval.) That is, take the difference between these two non-parametric plots, and plot them out.
- (iv) Try re-estimating (iii) for some smaller and larger bandwidths. Show at least one result that you think is “just barely too small a bandwidth” and one result that you

- think is “just barely too large a bandwidth”. Also show results that you think are most informative for answering the question “how does treatment impact depend on baseline test score”?
- (v) How similar or different are the impressions/punchlines for your preferred estimate, compared to your best “replication effort” estimate?
 - (vi) Based on your preferred estimate in (iv), what is your take on the impact of this program on girls whose baseline test scores were low? And more broadly across the distribution of baseline test scores, how does the impact vary?

3. Predicting House Prices

This problem builds on the data used in the JEP article by Mullainathan and Spiess, <https://www.aeaweb.org/articles?id=10.1257/jep.31.2.87> . The replication package link is at: <https://www.openicpsr.org/openicpsr/project/113993/version/V1/view>

The replication package appears to be “code only”, and I don’t see the data that they use. In their online appendix (<https://www.aeaweb.org/content/file?id=4212>), section B, they describe how they create their sample. You can use a similar process – you will have a different sample partition of “train” and “hold out” than they do. When you randomly sort the data to split into “train” and “hold out”, make sure to set a random number seed first, so that you can replicate your results.

I think that the microdata for the 2011 AHS can be downloaded from here: <https://www.census.gov/programs-surveys/ahs/data/2011/ahs-2011-public-use-file-puf/ahs-2011-national-public-use-file-puf.html> . I *think* (but I’m not yet 100% sure) that we want to work with the PUF National Flat File.

- a. Try to reproduce the sample selection and data cleaning, as described in Appendix section B1. How close do you get? Based on this exercise, do you have any advice for the authors on “best practice for reproducibility and replication”?
- b. Start by trying to replicate the OLS row of Table 1. NOTE: they indicate that their LHS variable is $\ln(\text{prices})$. You don’t have to replicate the Confidence intervals for the R^2 ; but if you think you are up to it, it’s good to try to do so.
- c. Add in a measure of RMSE. Add in the raw standard deviation of (log) house prices, as a comparison. How should we interpret the RMSE, given the data transformation on the LHS variable?
- d. Try to create a “kitchen sink” OLS model, using splines, polynomials, dummies, and interactions, that does a good bit better. Report out on how it does, in parallel to (a)-(c) above.
- e. Compare how your “kitchen sink” model does by quintile of house price, compared to the basic OLS model, in parallel to Table 1.
- f. Let’s see how good our models do across the distribution of $\hat{y}$. In the hold-out data, compute “squared prediction error” for each observation. Then use a local-linear nonparametric regression to estimate MSE, as a function of $\hat{y}$. Then take

the square root of that to have an estimate of RMSE as a function of $\hat{y}$. Then plot this out, focusing on the relevant values of your $\hat{y}$ (say, 1st percentile to 99th percentile). You can also plot out a horizontal line at the global RMSE. You might have to try different bandwidths to get a graph that you think best represents what is going on. You can just report your preferred bandwidth and its graph, in your answers. How does the OLS prediction model fit, across the distribution?

An upcoming theme in our class is that the distribution of the LHS variable can impact our choices, and the performance of our models. I suspect that the house-price distribution is Right-skewed.

- g. Using a histogram or kernel density, show the distribution of “raw” (untransformed) housing prices. Compute summary statistics, including the skewness statistic. Does it look like prices are skewed?
- h. Suppose now we are interested in the goodness of fit for “levels of house prices”. One approach is to take our predictions (from parts (b) and (d)) for “log prices”, and then exponentiate the predictions, to get predictions in levels. Use this approach to re-compute the R^2 and RMSE (and raw SD comparisons) measures in your training and hold-out samples. Do this for both your “replication” model and your “kitchen sink” model.
- i. Now, using the same set of features/covariates, re-run your OLS models with “level of house price” as the LHS variable. And re-compute the R^2 and RMSE for both data sets. How do the results compare to your results in (g)? Which approach does better?